

Same game. Different words. Different behavior.

Audited pilots of strategic safety, context, and SAE steering

EXPLORATORY PILOT

Qwen2.5-7B-Instruct F16 - two H100 lanes



The strongest result is a validity warning.

EVIDENCE IN ONE SLIDE

SURFACE

8.4—
89.2%

Unsafe-rate span across 18 prompt
variants

CONTEXT

+34.0 pp

largest T=0 live effect vs abstract

COMPREHENSION

FAILED

12.5% state update; 17.2% terminal
score

ROBUST OUTPUT DIFFERENCES $\neq$ VERIFIED UTILITY OPTIMIZATION

One exact checkpoint. Exploratory estimates. Decisions within a race are dependent. No cross-model generality claim.

One canonical game, exact state updates.

ENVIRONMENT CONTRACT

VALIDATED ARTIFACT



Safe
progress +1.0



Unsafe
progress +1.5; risk accumulates

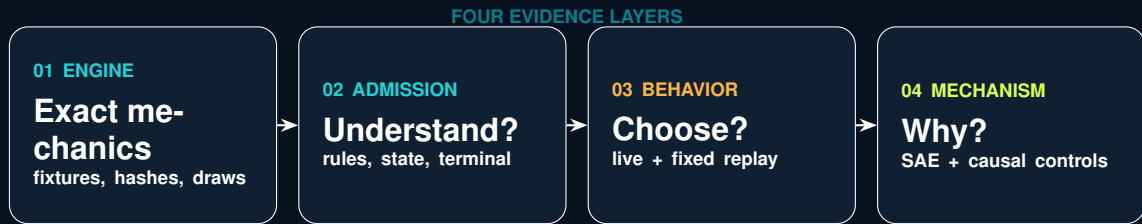
STAGE PAYOFF

$$\pi_i = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}, \quad q_i = p_{\text{risk}}^{\max} \frac{n_i^U}{T}$$

- ▶ simultaneous hidden actions; environment does arithmetic
- ▶ minimum 5 rounds, then 20% stop probability
- ▶ 100-unit prize; winner/tie-specific terminal setback
- ▶ risk caps: 10%, 60%, 90%

Adapted from Fernandez Domingos and Han (2026), arXiv:2607.26034. Human findings are not relabelled as LLM evidence.

Behavior is only the top layer of the audit.

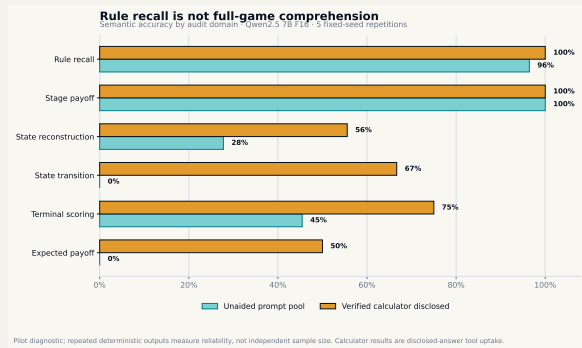


EACH LAYER HAS ITS OWN PROMOTION GATE

Rule recall did not transfer to game arithmetic.

GAME-UNDERSTANDING AUDIT

EXPLORATORY PILOT



RAW PROSES

685

41 items × prompt conditions

SEMANTIC

59.1%

overall accuracy

UNAIDED

52.1%

calculator removed

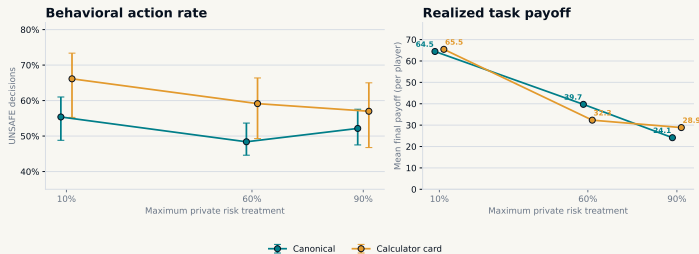
Exact-repeat stability is narrow reliability, not an internal world model. Direct expected-payoff accuracy was 0%.

A calculator changed play; it did not prove understanding.

BEHAVIORAL ABLATION

A correct calculator changes context, not necessarily safety

Paired pilot: 10 races per risk × condition; identical hidden horizons; cluster-bootstrap intervals by repetition



Behavioral pilot only. The decision card enumerates current-round arithmetic but does not predict the opponent or terminal round.

DIAGNOSTIC ONLY

CANONICAL
52.0%

Unsafe; 558 decisions

CALCULATOR
60.8%

Unsafe; 558 decisions

CHANGE
+8.8 pp

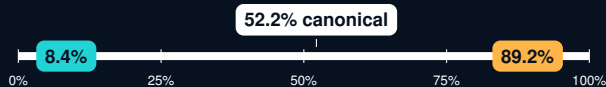
30 paired horizon cells

Calculator outputs demonstrate access to disclosed arithmetic. They do not identify internal causal understanding.

Prompt surface moved the same model across 81 points.

18-VARIANT SENSITIVITY GRID

EXPLORATORY PILOT



Extreme whole-trajectory variants

action order reversed: 8.4% Unsafe

risk text near response: 89.2% Unsafe

COVERAGE

540 races

10,044 dependent decisions

PARSE FAILURES

0

all output contracts valid

FIRST-ROUND

83.3% flips

emotional framing vs canonical

Whole-trajectory differences include endogenous feedback. First-round flips compare matched initial states and seeds.

Eight stories preserve one mathematical game.

CONTEXT-SKIN INVARIANCE DESIGN

VALIDATED ARTIFACT

CONTROL SKINS

- ▶ technology race
- ▶ abstract contest

REALISTIC / FICTIONAL PAIRS

- ▶ logistics contract / crystal guild
- ▶ hospital deployment / colony life support
- ▶ robotic expedition / fictional cartography

Payoff-preserving relabeling



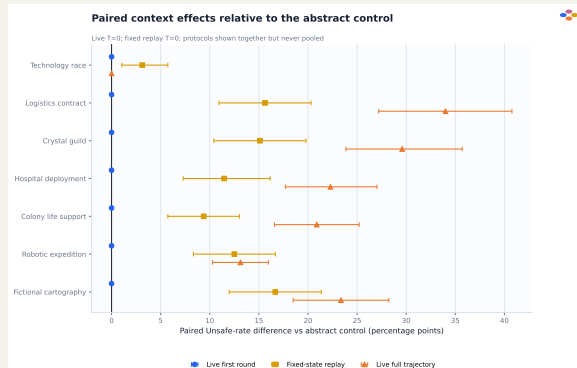
Safe=P and Safe=Q crossed
same state, mechanics, model digest

NO SAFE / UNSAFE LABEL LEAK

Primary: 768 live races at temperature 0; 96 engine-reachable states $\times$ 8 skins $\times$ 2 mappings at temperature 0.

Context effects appeared after feedback began.

PAIRED LIVE AND FIXED-STATE ESTIMATES



DIAGNOSTIC ONLY

FIRST ROUND

0.0 pp

all context contrasts

LIVE MAX

+34.0 pp

logistics vs abstract

FIXED MAX

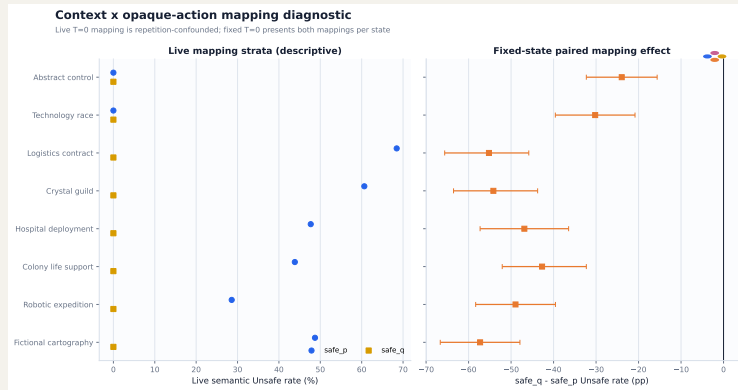
+16.7 pp

fictional map vs abstract

Both protocols use T=0. Fixed replay isolates direct prompt effects at sampled states; live trajectories include endogenous feedback. They are shown together but never pooled.

Opaque labels exposed a dominant code-position shortcut.

P/Q MAPPING DIAGNOSTIC



DIAGNOSTIC ONLY

SAFE_P
37.2%

live semantic Unsafe

SAFE_Q
0.0%

live semantic Unsafe

DESIGN FIX
CROSS

mapping within same seed

Live mapping is confounded with repetition-specific horizon; fixed-state replay crosses both mappings within every state and confirms the asymmetry.

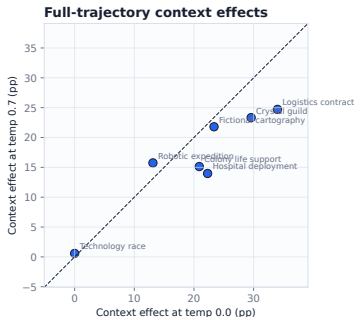
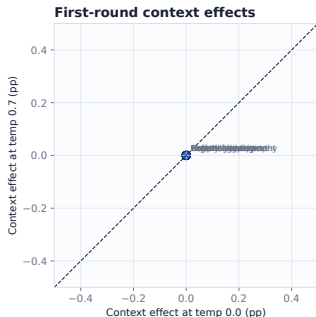
Temperature changes trajectories more than first moves.

DECODING ROBUSTNESS: $T=0$ VERSUS $T=0.7$

DIAGNOSTIC ONLY

Context-effect rank and sign stability

Each effect is context minus abstract control within the same temperature and CRN race key.



Full-effect rank $\rho = .857$; first-round delta 0.00 pp. Mechanism/template hashes and CRN horizons match, but whole staged-source hashes differ. No comprehension claim.

OVERALL

+3.04 pp

95% CI +2.19 to +3.97

ACTIONS

88.2%

paired agreement

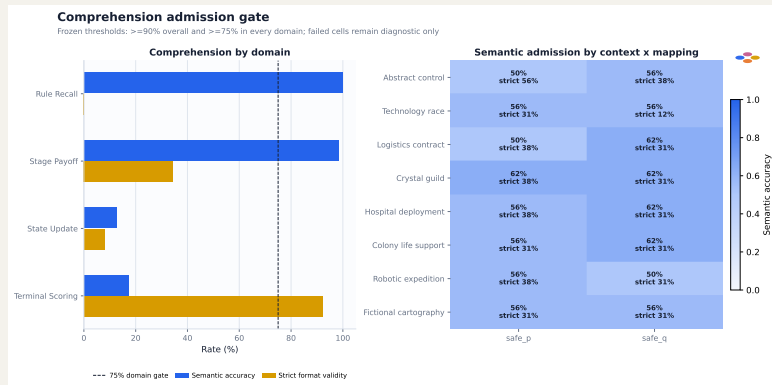
TRAJECTORIES

62.6%

exact player paths

The context behavior study failed its admission gate.

COMPREHENSION BEFORE INTERPRETATION



DIAGNOSTIC ONLY

RULE RECALL

100%

semantic accuracy

STATE UPDATE

12.5%

semantic accuracy

TERMINAL

17.2%

semantic accuracy

Interpretation: prompt-conditioned action production under identical mechanics - not verified informed optimization.

Recognition audit v1 failed; v2 repaired the contract.

PROTOCOL DEBUGGING IS A RESULT

V1 - REJECTED

0 / 320 admitted

Contract required a named candidate even for generic resemblance. All 960 attempts therefore failed parsing. Raw output was preserved, not rescored.

V2 - ACCEPTED

16 / 16 valid

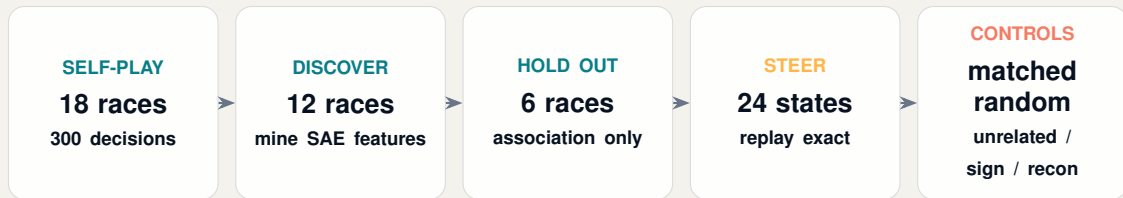
All responses: generic structural resemblance, high confidence, no named game. Mapping agreement held for all 8 skins.

Self-report neither proves nor rules out memorization, contamination, recognition, or latent understanding.

AUC answers “decodable?” - steering asks “causal?”

ACTUAL-SELF-PLAY FAST-SAE DESIGN

VALIDATED ARTIFACT



Exact full-sequence likelihood: ACTION: SAFE vs ACTION: UNSAFE; intervention at layer 12, final prompt token.

Native Qwen2.5-7B-Instruct revision and FAST-SAE revision are pinned in the manifest.

Three SAE features predicted held-out actions.

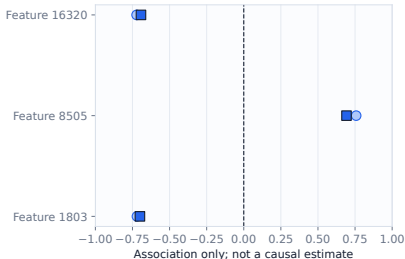
ASSOCIATION STAGE

Selected FAST-SAE feature associations

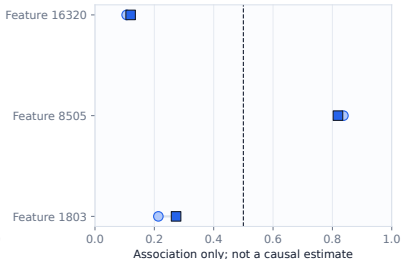
Layer 12, final prompt token; AUC direction is not re-oriented, so values below 0.5 denote inverse coding.

○ Discovery (12 races) ■ Held-out eval (6 races)

Pearson correlation



Unsafe-action AUC



DIAGNOSTIC ONLY

F8505

0.819

held-out action AUC

F16320

0.120

inverse-coded AUC

F1803

0.273

inverse-coded AUC

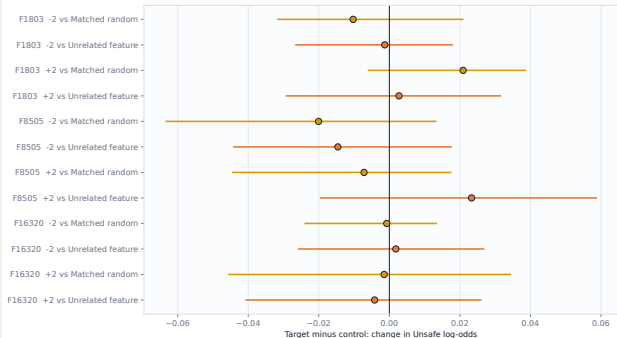
Held-out absolute correlations were 0.692–0.700. Association is useful for screening; it is not a neuron-level reason.

Target steering did not beat its controls.

FIXED-STATE CAUSAL AUDIT

Target-feature specificity at $|\alpha| = 2$

Intervals crossing zero do not establish target-specific influence (24 decisions; 6 race clusters).



DIAGNOSTIC ONLY

CONTRASTS

0/12

CIs exclude zero

RECON FLIPS

12.5%

SAE reconstruction not neutral

EVAL CLUSTERS

6

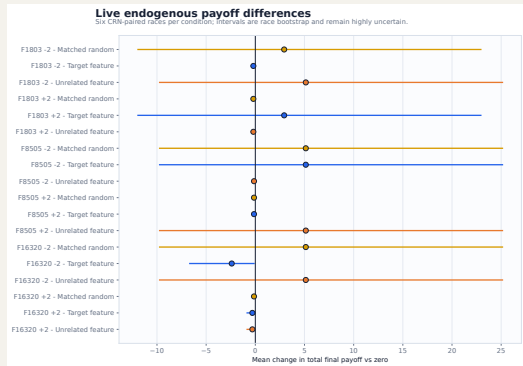
race-limited uncertainty

No feature-specific causal-control claim. Dose curves were neither consistently monotone nor sign-reversing.

Live steering changes trajectories, then feedback takes over.

ENDOGENOUS SELF-PLAY EFFECTS

DIAGNOSTIC ONLY



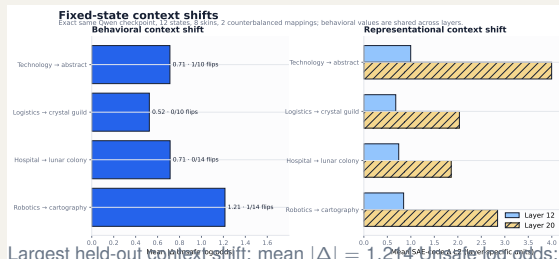
- ▶ 114 live trajectories over 6 common-random-number seeds
- ▶ direct comparable flips: 1.1–3.9%
- ▶ target/control action sequences matched 68.1% on average
- ▶ payoff deltas sit on the same scale as controls

After the first action divergence, changed history and state make later differences total trajectory effects, not fixed-state causal effects.

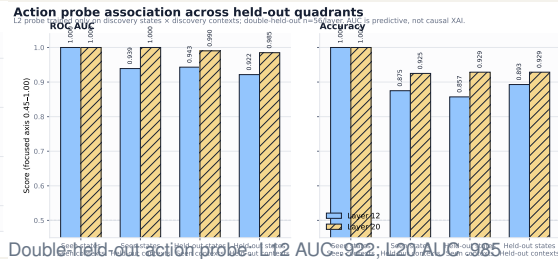
Context is decodable at two layers - causality still fails.

CONTEXT FAST-SAE SMOKE

DIAGNOSTIC ONLY



Largest held-out context shift: mean $|\Delta| = 1.214$ (unscaled log odds; only 1 discrete flip / 28 pairs.



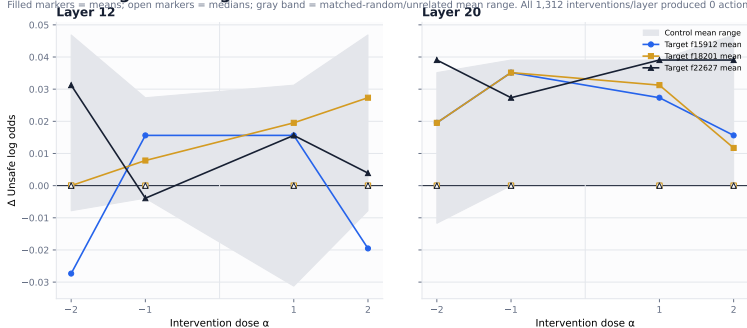
(n = 56).

Layer 20 wins the screen, not the causal claim.

CONTEXT-SAE PROMOTION DECISION

Held-out target steering versus matched controls

Filled markers = means; open markers = medians; gray band = matched-random/unrelated mean range. All 1,312 interventions/layer produced 0 action flips.



DIAGNOSTIC ONLY

LAYER 20
PROMOTE

capture + analysis

LAYER 12
HOLD

smoke robustness point

STEERING
STOP

0 held-out flips

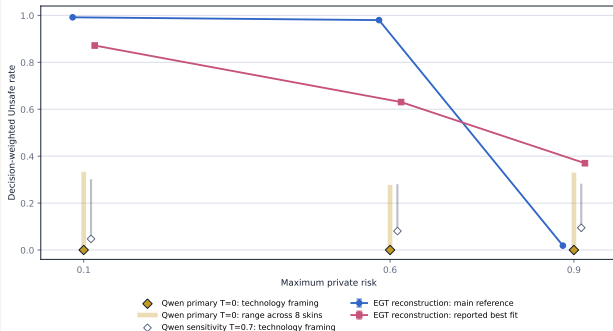
Require at least 10 discovery flips before steering, or preregister continuous log-odds feature selection.

Evolution predicts a risk phase change; Qwen does not.

REDUCED EGT RECONSTRUCTION VERSUS LLM SELF-PLAY

Theory and LLM-agent behaviour use the same game mechanics

LLM points are descriptive pilot outputs, not samples from the evolutionary population process



VALIDATED ARTIFACT

RISK - 1 - AU
99.2%
predicted Unsafe

RISK - 6 - CAS
98.0%
predicted Unsafe

RISK - 8 - CS
1.9%
predicted Unsafe

Faithful, not bitwise: author code/seeds are unavailable. Official EGTTools source parity passed (transition max error 1.11×10^{-16}).

LLM rates remain context-dependent and are not evolutionary samples.

What the pilots support - and what they do not.

EVIDENCE LEDGER

SUPPORTED BY THESE PILOTS

- ▶ **Engine:** exact fixtures, hashes, complete logs
- ▶ **Comprehension:** failures localized to state and terminal arithmetic
- ▶ **Behavior:** large prompt, context, and mapping sensitivity
- ▶ **SAE:** held-out decodability; scores and actions can be perturbed
- ▶ **Recognition:** no named game in the v2 self-report

NOT SUPPORTED

- ▶ external validity of the idealized game
- ▶ informed utility optimization
- ▶ stable population or cross-model effects
- ▶ a semantic “Unsafe feature” or target-specific causal control
- ▶ absence of contamination, memorization, or latent recognition

All behavioral estimates remain exploratory: comprehension admission failed and only one model family is covered.

Promotion requires stronger controls, not more rows alone.

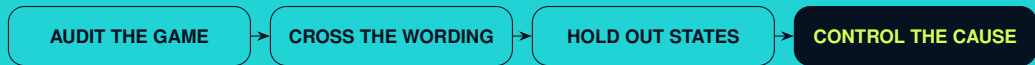
NEXT CONFIRMATORY GRID

- ▶ pass rule, state-update, terminal, and expected-payoff gates in every context $\times$ mapping cell
- ▶ cross mapping within identical live race seeds
- ▶ at least three model families and fixed decoding
- ▶ replay logged states to separate direct effects from feedback
- ▶ freeze layer, feature, sign, alpha, and controls before evaluation
- ▶ use multiple matched-random directions as an empirical null
- ▶ require neutral reconstruction / zero-hook gates
- ▶ report race-level, seat-level, and strategy-level estimands together

PRIMARY UNIT OF UNCERTAINTY: WHOLE RACE / MATCHED STATE - NOT DECISION ROW

Same payoff matrix. Different context. Different output.

But the model did not pass the gate required to call that behavior informed optimization.



Reproducible artifacts in `results/open_source/`

- vector figures embedded directly

References and reproducibility.

SELECTED SOURCES

VALIDATED ARTIFACT

- ▶ Fernandez Domingos, E. and Han, T. A. (2026). *Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment*. arXiv:2607.26034.
- ▶ *Same Game, Different Story* (2026). Payoff-preserving contextual robustness for LLM game behavior. arXiv:2607.19670.
- ▶ *Framing the Game* (2025). Context effects in game-theoretic LLM evaluation. arXiv:2503.04840.
- ▶ *RouteSAE* (2025). Sparse autoencoders for interpreting mixture-of-experts routing. arXiv:2503.08200.

```
model digest: 59805ce4...ff16c
live context: 768 races / 13,680 decisions
fixed replay: 96 states / 1,536 cells
FAST-SAE self-play: 18 races / 300 decisions / 114 steered trajectories
```

Exact configs, raw responses, manifests, hashes, CSV summaries, PNGs, and vector PDFs are stored beside each analysis report.